SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA1741**

Register Number	**192419033**
Name	**VISHAL.V**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS:2 **Total Marks:20**

Descriptive Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behaviour and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach wou ld be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.

   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Code of Conduct:

I VISHAL.V(192419033) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: VISHAL.V

RUBRICS FOR CALCULATION:

Criterion	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions	5
Explanation	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation	5
Application	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis	4
Organization	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers	3
Accuracy	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps	2
Clarity	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand	1

SOLUTIONS: -

Answer 1: Intelligent Agents in a Smart Home Automation System

Introduction:
An intelligent agent is a system that observes its surroundings and takes suitable actions. In a smart home, it helps control lights, fans, temperature and security automatically based on the user's needs and environmental conditions.

Types of Intelligent Agents:

1. Simple Reflex Agent: -
A simple reflex agent takes action based on the current situation.
For example, when motion is detected, it switches ON the light. When the temperature increases, it switches ON the fan.
Its main advantage is that it works quickly and is easy to design.
However, it cannot remember past events or learn user preferences.

2. Model-Based Agent: -
A model-based agent remembers information about its surroundings and uses it to make decisions.
For example, it can remember whether someone is inside a room and control the lights accordingly. It works well when the environment changes, but it requires memory to store information.

3. Goal-Based Agent: -
A goal-based agent works towards achieving a particular goal.
For example, it tries to maintain a comfortable room temperature and ensures that doors are locked when nobody is at home. It makes decisions based on the desired result, but it may require more time to plan its actions.

4. Utility-Based Agent: -
A utility-based agent chooses the action that provides the best overall benefit.

For example, it balances room comfort and electricity consumption by adjusting the air conditioner. It helps make better decisions, but choosing the best balance between different needs can be difficult.

Best Agent for a Smart Home:
A hybrid agent combining model-based, goal-based and utility-based approaches would be the most suitable. It can remember room conditions, work towards comfort and security, and reduce electricity usage. It can also learn the user's preferences over time.

Conclusion:
Intelligent agents make smart homes safer, more comfortable and energy-efficient. A hybrid approach is the best choice because it combines memory, goal achievement and good decision-making.

Answer 2: Uninformed Search Strategies for Solving a Maze

Introduction:
Uninformed search is a method of finding a solution without knowing the best path in advance. A robot placed in an unknown maze can use these search methods to explore different paths and find the exit.

1. Uniform Cost Search (UCS): -

Uniform Cost Search explores the path with the lowest total cost first.
Working:

1. The robot starts from the initial position.
2. It selects the position with the lowest path cost.
3. It checks the neighbouring positions and adds their costs.
4. It continues until the exit is found.

Advantages:

- Finds the least-cost path when movement costs are positive.
- Useful when different paths have different costs.

Disadvantages:

- Uses more memory to store possible paths.
- Can take more time in a large maze.

2. Iterative Deepening Search (IDS): -

Iterative Deepening Search explores the maze using increasing depth limits until it finds the exit.
Working:

1. The robot starts searching with a depth limit of zero.
2. It searches all positions within that limit.
3. If the exit is not found, it increases the limit.
4. It repeats the process until the exit is reached.

Advantages:

- Requires less memory.
- Finds the shallowest solution when all moves have equal cost.

Disadvantages:

- Repeats some searches.
- May not find the least-cost path when different moves have different costs.

Comparison of UCS and IDS

Feature	UCS	IDS
Search method	Lowest-cost path first	Increasing depth limits
Time complexity	Can be exponential	\(O(b^d)\)
Space complexity	Can be exponential	\(O(bd)\)
Best use	Different movement costs	Equal movement costs

Here, b is the number of possible next moves and \(d\) is the depth of the solution.

Role of Intelligent Agents:
Different intelligent agents help the robot make decisions in the maze.

- Simple reflex agent: Reacts to nearby obstacles but may repeat paths.
- Model-based agent: Remembers visited positions and obstacles.
- Goal-based agent: Chooses actions to reach the exit.
- Utility-based agent: Considers time, distance and energy while selecting a path.

Best Combination

A combination of a model-based agent, a goal-based agent and UCS would be suitable when different paths have different costs. The model-based agent remembers the maze, the goal-based agent focuses on reaching the exit, and UCS helps find the least-cost path.
If all moves have equal costs and memory is limited, IDS would be a better choice.

Conclusion

UCS is suitable when finding the least-cost path is important, while IDS is useful when memory is limited. Choosing the right search method and intelligent agent helps the robot find the exit more effectively.